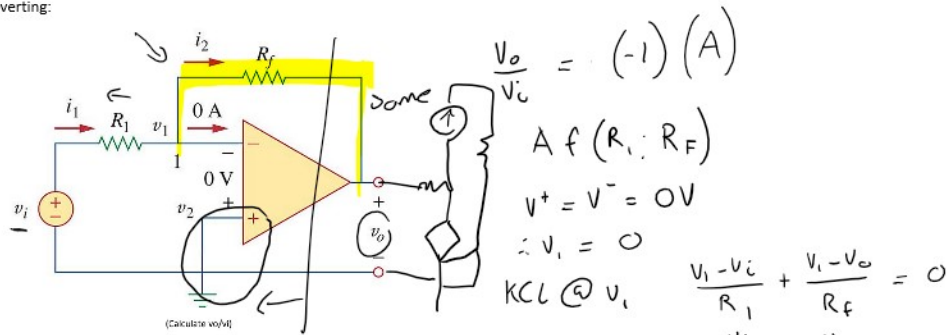


L19 - Inverting and Non-inverting Amplifiers

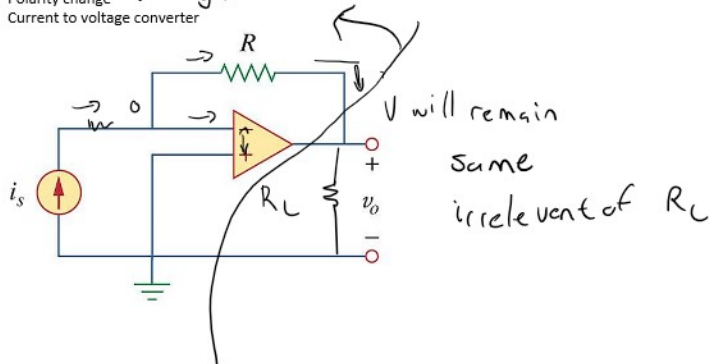
Thursday, March 14, 2024 11:10 AM

Inverting:

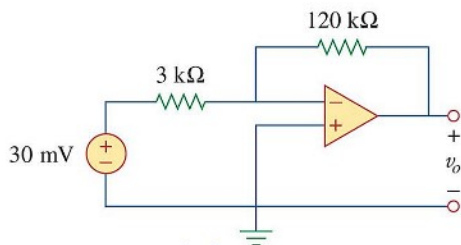


Use cases:

- Polarity change with gain
- Current to voltage converter



Example:

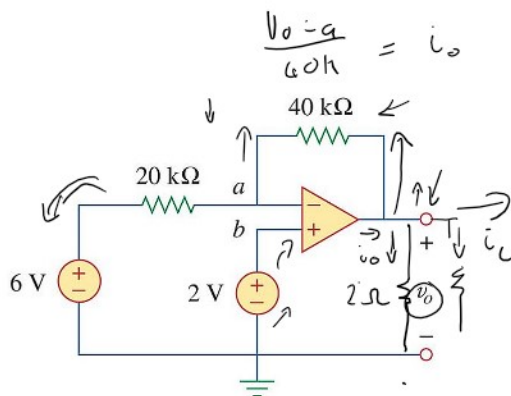
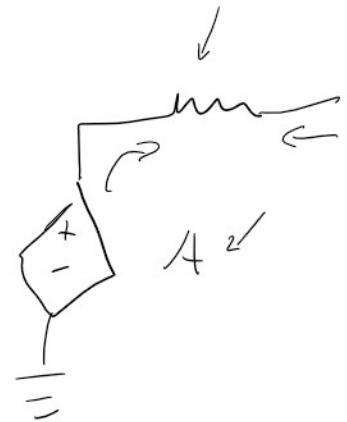


$$v_o = v_i \left(-\frac{R_f}{R_i} \right)$$

$$= 30m \left(-\frac{120k}{30k} \right)$$

$$= 30m (-4)$$

$$= -120mV$$



$$v^+ = v^- = 2V$$

$$v_a = 2V$$

$$KCL @ \text{node } a$$

$$\frac{2-6}{20k} + \frac{2-v_o}{40k} = 0$$

$$2(-4) + 2 - v_o = 0$$

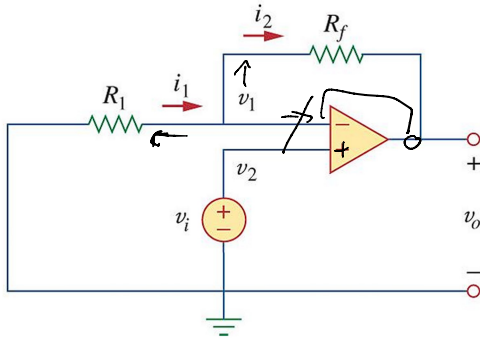
$$v_o = -6V$$

$$\therefore i_o = -\left(\frac{2 - v_o}{40k} \right)$$

$$= \frac{8}{40k}$$

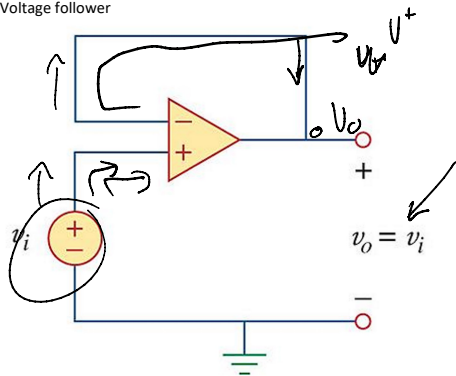
$$= 0.2mA$$

Non-inverting:

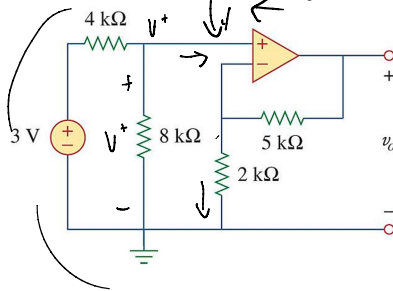


Use cases:

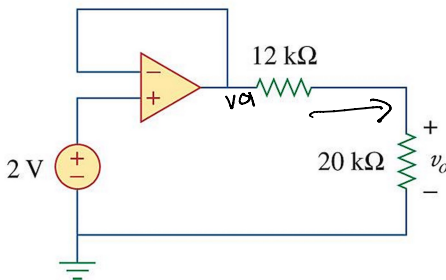
- Change positive gain
- Voltage follower



Example:



Example:



$$\frac{v_o}{v_i} = A$$

$$v^+ = v^- = v_i$$

KCL @ v^-

$$\frac{v_i - 0}{R_i} + \frac{v_i - v_o}{R_f} = 0$$

$$v^+ = v^- = v_i \quad v_i \left(\frac{1}{R_i} \right) + v_i \left(\frac{1}{R_f} \right) + v_o \left(\frac{-1}{R_f} \right) = 0$$

$$v_i \left(\frac{1}{R_i} + \frac{1}{R_f} \right) + v_o \left(\frac{-1}{R_f} \right) = 0$$

$$\frac{v_o}{v_i} = \left(\frac{-\frac{1}{R_f}}{\frac{1}{R_i} + \frac{1}{R_f}} \right)$$

$$v^+ = 3 \left(\frac{8k}{4k + 8k} \right)$$

$$= 2V$$

$$v^+ = v^- = 2V$$

KCL @ v^-

$$\frac{2 - 0}{2k} + \frac{2 - v_o}{5k} = 0$$

$$v^+ = v^- = 2V$$

$$\therefore v_a = 2V$$

$$v_o = v_a \left(\frac{20k}{12k + 20k} \right) = 2 \left(\frac{20}{32} \right)$$

